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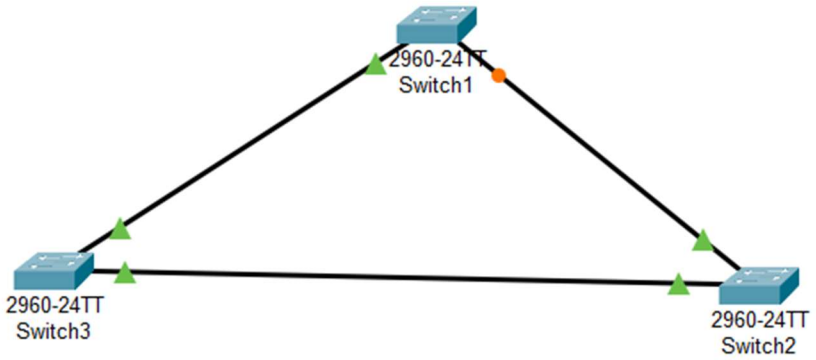
Theoretical preparation

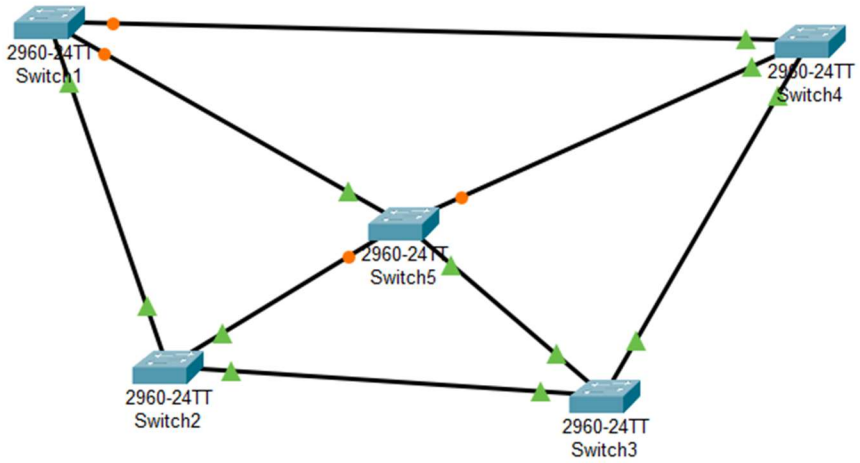
The Spanning Tree Protocol (STP) is a connection management protocol responsible for ensuring the elimination of loops at the local network level when multiple switches are connected to each other. The switches communicate with each other through Bridge Protocol Data Units (BPDUs). There are three kinds of messages: Configuration BPDU, Topology Change Notification BPDU, and Topology Notification Acknowledgement BPDU. STP elects a Root Switch based on the Bridge Priority parameter. Furthermore, ports are defined in four categories: Root Ports, Designated Ports, Alternative Ports, and Non-Designated Ports.

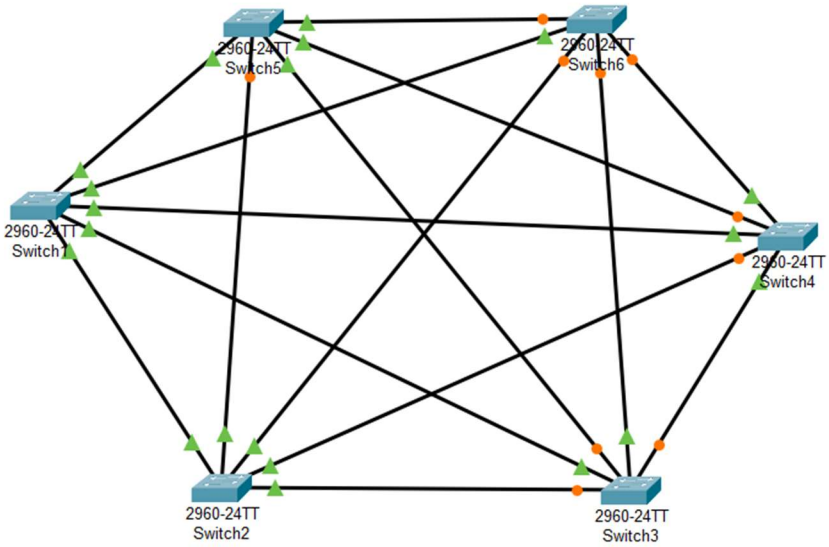
Course of work

The workflow began with creating an initial network to understand the fundamental operations of the STP protocol. Subsequently, we designed two larger network topologies, where we manually changed the Root Bridges and adjusted Bridge Priorities. Overall, the process was straightforward as the instructions were clear and concise.

Screenshots






2960-24TT
Switch5
2960-24TT
Switch6
2960-24TT
Switch1
2960-24TT
Switch4
2960-24TT
Switch2
2960-24TT
Switch3

Answers to the questions

Step 1

Are all indications green now? If not, why?

As observed in the network diagram, not all indicators are green. The protocol has blocked the link between Switch 1 and Switch 3 to prevent a broadcast storm and maintain a loop-free topology.

In the CLI on each Switch, run the command show spanning-tree vlan 1

```
Switch>show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97D1.9D71
             Cost        19
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00D0.5859.6EC5
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/2            Root FWD 19        128.2    P2p
Fa0/1            Altn BLK 19        128.1    P2p

Switch>
```

Switch 1

```
Switch>show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97D1.9D71
             Cost        19
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000C.CF26.278B
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/2            Root FWD 19        128.2    P2p
Fa0/1            Desg FWD 19        128.1    P2p

Switch>
```

Switch 2

```
Switch>show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0001.97D1.9D71
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0001.97D1.9D71
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/1            Desg FWD 19        128.1    P2p

Switch>
```

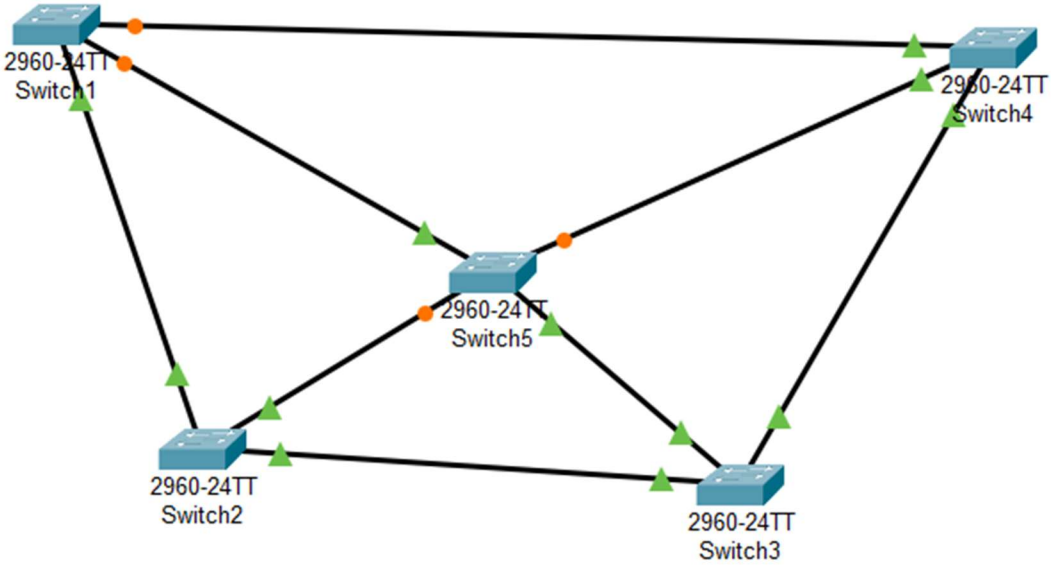
Switch 3

Which Switch is the Root, which port has been blocked, and why?

Switch 3 was elected as the Root Bridge, as indicated by the fact that the ports on the other two switches leading to it are designated as Root Ports. Port 1 on Switch 1 has been placed in a Blocking state to eliminate the loop.

Step 2&3

Check if there is a closed loop where all indicators are green. (screenshot)



There is no loop where all indicators remain green. Opening a blocked port would immediately result in a network loop, which STP is designed to prevent.

Check if any port could be "green" without a loop

It is not possible to transition a port to a forwarding state (green) without creating a loop.

In the CLI on each Switch, run the command (screenshot)

Switch>show spanning-tree vlan 1

```
Switch>show sp vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    12289
             Address     0006.2A09.3D4E
             Cost        38
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577  (priority 24576 sys-id-ext 1)
             Address     0002.17C1.A16C
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Altn BLK 19        128.2    P2p
Fa0/3            Altn BLK 19        128.3    P2p

Switch>
```

Switch 1

```
Switch>show sp vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    12289
             Address     0006.2A09.3D4E
             Cost        19
             Port        2(FastEthernet0/2)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0001.C9ED.8B3E
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Root FWD 19        128.2    P2p

Switch>
```

Switch 2

```
Switch>show sp vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    12289
             Address     0006.2A09.3D4E
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    12289  (priority 12288 sys-id-ext 1)
             Address     0006.2A09.3D4E
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/3            Desg FWD 19        128.3    P2p

Switch>
```

Switch 3

```
Switch>show sp vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    12289
             Address     0006.2A09.3D4E
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     000C.8516.CBA8
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p

Switch>
```

Switch 4

```
Switch>show sp vlan 1
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    12289
             Address     0006.2A09.3D4E
             Cost        19
             Port        4(FastEthernet0/4)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00D0.9731.4D4E
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Altn BLK 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/3            Altn BLK 19        128.3    P2p
Fa0/4            Root FWD 19        128.4    P2p

Switch>
```

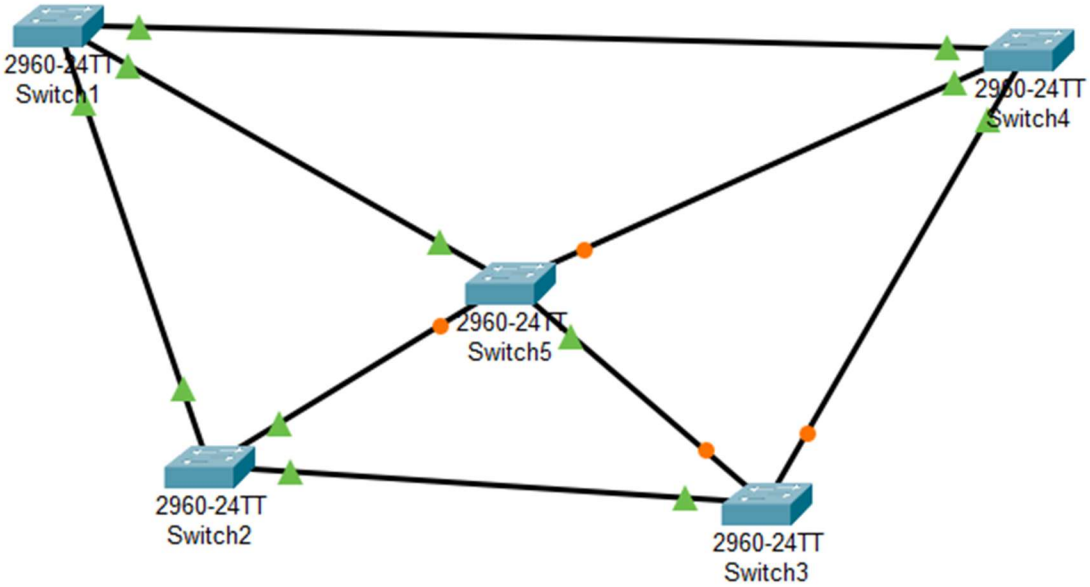
Switch 5

Which switch is elected as Root and which ports are blocked?

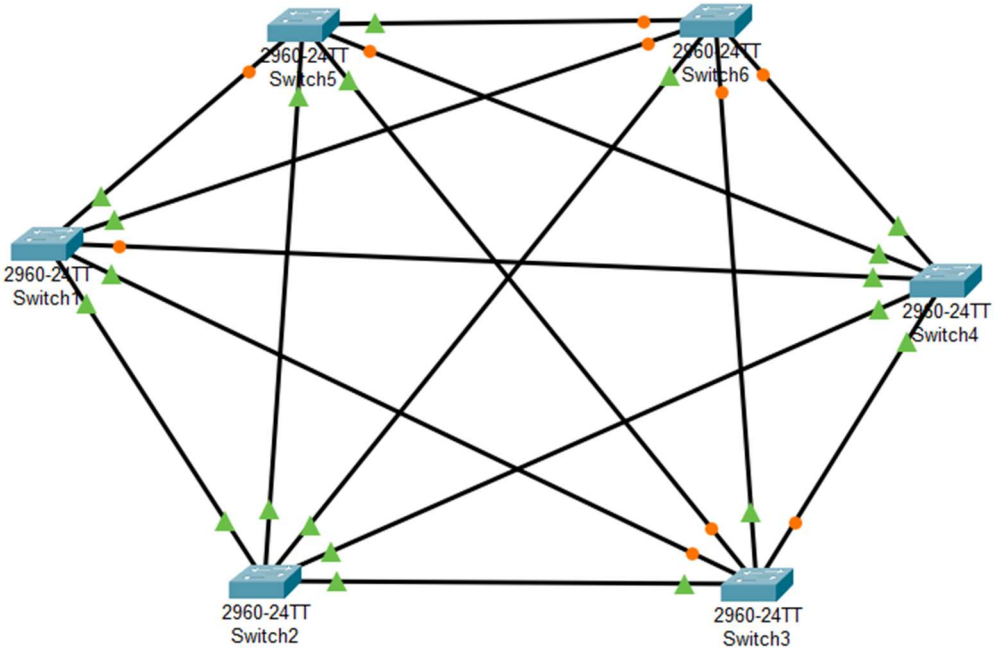
Switch 3 remains the root as all ports leading directly to it are active Root Ports. Redundant ports are set to Alternate (Blocking) status to ensure only one active path exists between any two segments.

What has changed in the network and why? (screenshot)

Which Switch is now the root? (screenshot)



After manually configuring **Switch 1** as the root, all of its ports transitioned to a **Designated Forwarding (green)** state. The other switches in the network re-calculated their path costs and blocked different ports to adapt to the new topology.

Step 4

Configure the STP protocol so that all ports of S2 and S4 are FWD (green forward). (screenshot)